Day 7 Assignments

Write a C++ program using inheritance, runtime polymorphism and templates to calculate parking charges for different types of vehicles.

Parking Types

Vehicle Type: Car

Fixed parking charge: Rs. 100 for first 3 hours

Beyond 3 hours: Rs. 30 per hour

No discount available.

Vehicle Type: Truck

Fixed parking charge: Rs. 200 for first 2 hours

Beyond 2 hours: Rs. 50 per hour

Discount available: Code TRUCK50 gives Rs. 50 off

Program Requirements

Use abstract base class Vehicle with a pure virtual function calculateCharge().

Create derived classes Car and Truck.

Use runtime polymorphism to handle both vehicle types dynamically.

Implement a static data member to count how many parking receipts are generated.

Use a function template to validate the discount code.

Use an inline function to compute extra parking charges.

Input Validation (Exception handling)

Parking hours must be positive.

If vehicle type is invalid, prompt again.

Sample Input / Output

Sample Input 1

Enter type of Vehicle (C for Car, T for Truck): C

Enter parking hours: 5

Expected Output

Vehicle type = Car

Extra charge = Rs. 60

Total Parking Charge = Rs. 160

Receipts generated so far: 1

Sample Input 2

Enter type of Vehicle (C for Car, T for Truck): T

Enter parking hours: 5

Enter discount code if available (or press Enter to skip): TRUCK50

Expected Output

Vehicle type = Truck

Extra charge = Rs. 150

Total Charge after discount = Rs. 300

Receipts generated so far: 2